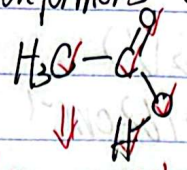
Group A 2026/2/3 Laboratory #10 Computational Chemistry

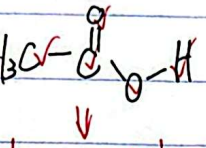
By Yuchen Shi with Group members: Siqi Li & Yizhe Dai

Part I: Build Conformers of Acetic Acid & Minimize

Conformer 2: 

{ Final Strain Energy: −85.9140 kJ/mol
{ Symmetry Group: Cs

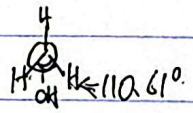
Atoms occupying the plane

Conformer 1: 

{ Final Strain Energy: −110.4824 kJ/mol
{ Symmetry Group: Cs

Atoms occupying the plane

* Table. 1 Calculated Parameters of Acetic Acid

		Conformer 1 (more stable)	Conformer 2
	Dihedral Angle (H-O-C=O)	0.00°	0.00°
	Strain Energy (kcal/mol)	−110.4824	−85.9140
	Point Group	Cs	Cs
	Symmetry element	sp^3 C. O. O-H (E. σ)	sp^3 C. O. O-H (E. σ)
	Conformation	Eclipse	Eclipse
HF/3-21G ⇐	Dipole Moment	1.65 Debye	4.97 Debye
	Methyl Group H-C-H angle	108.63°/108.63°/110.61° 	108.55°/108.55°/110.70°
Extra data from HF/3-21G	Energy (a.u.)	−226.534237	−226.521183
	Expt. Heat (kJ/mol)	−434.310	−434.300
	E LUMO (eV)	5.09	5.05
	E HOMO (eV)	−12.02	−11.91

Part II. Calculations of Formaldehyde

(1) Optimize Energy : 0.2266 kJ/mol (on MM)

(2) # of vibrational modes: 6 (linear, symmetry C_{2V})

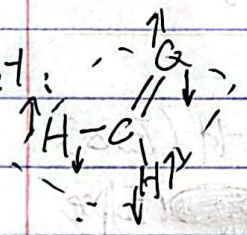
(3) A_2 { CH_2 symmetric stretching: 3162 cm^{-1}
CO stretching: 1915 cm^{-1}
CH_2 scissoring: 1693 cm^{-1}

B_1 { CH_2 asymmetric stretch: 3233 cm^{-1}
CH_2 rocking: 1379 cm^{-1}

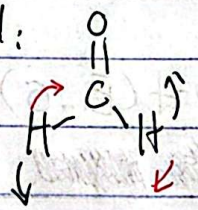
B_2: Out of plane bend: 1337 cm^{-1}

(4) HOMO Energy: -11.78 eV
LUMO Energy: 4.01 eV
LUMO+1 Energy: 7.6 eV
HOMO-1 Energy: -14.3 eV

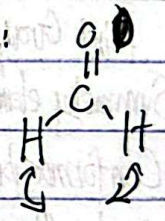
1337 cm^{-1}:



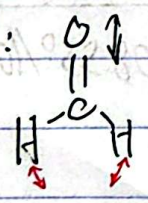
1379 cm^{-1}:



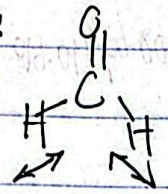
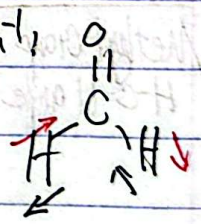
1693 cm^{-1}:



1915 cm^{-1}:



3162 cm^{-1}:



3233 cm^{-1}: